Asia Datathon Spring 2024 Report

Exploring the relationships of the carbonated beverages industry with teenage obesity and sugar consumption

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Contents

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1 Non-Technical Summary

1.1 Background

Main Questions

- What is the relationship between soda consumption and obesity rates across states and races?
- What is the statistical relationship between sugar prices, sugar imports and the carbonated beverages industry?

In the United States, the rising obesity rates pose a significant public health challenge, influenced by factors including dietary habits and socioeconomic, cultural, and environmental contexts. Notably, the consumption of sugar-sweetened beverages like soda is under scrutiny for its potential role in escalating obesity levels. This relationship, however, is multifaceted, varying across states and racial demographics, reflecting the diverse socio-economic landscapes and cultural practices that influence dietary choices and health outcomes.

Our study extends beyond the impact of soda on obesity to examine the intricate dynamics between sugar prices, sugar imports, and the carbonated beverages industry. The economic elements of sugar availability directly affect the beverage sector's production and pricing strategies, influencing consumption patterns. This analysis illuminates how global economic trends and trade policies indirectly shape dietary preferences and public health, emphasising the need for comprehensive strategies to address obesity, especially among lower-income groups most affected by the affordability of sugar-sweetened beverages.

Hence, a detailed investigation into the connections between soda consumption, sugar economics, and obesity rates is crucial. It offers insights into socio-economic and cultural influences on health behaviors, informing targeted public health interventions. This research aims not only to unravel the complexities of obesity in the U.S. but also to foster equitable health outcomes across its diverse population.

1.2 Summary of Findings

We found a significant two-way link between how much soda people drink and obesity levels across different states and racial groups in the United States. Using a detailed analysis method, we discovered that drinking soda and obesity rates influence each other, especially among teenagers. The more soda teenagers drink, the higher their risk of becoming obese, which in turn may change their consumption behaviour, causing a potential increase in soda consumption. Additionally, elevated obesity levels during a certain period could be associated with rising sales of soda and processed foods in preceding years. This trend may motivate soft drink manufacturers to boost production in the years that follow, providing another explanation for the bidirectional causality.

We also looked into how the production of meat and the performance of sugar-related companies in the stock market might affect this relationship. Our findings suggest that strategies to reduce soda drinking among young people could play a crucial role in tackling obesity.

The study provides important insights for those making policies and health programs, showing that focusing on reducing soda intake could help address the obesity crisis. Additionally, the stock market data we analyzed offer valuable information for investors and companies in the food and beverage industry.

2 Technical Exposition

2.1 General Approach

In our exploration of the data and the topic in general, we found the prevalence and bidirectional causality and positive feedback loops in data related to obesity, processed food consumption and other relevant variables. As a result, we decided to use variations of Vector Autoregressive models for our project, as they consider the joint dynamics and interactions between multiple variables, making it an ideal tool to capture reciprocal relationships.
The Vector Autoregression (VAR) model is a statistical model used in time series analysis to capture the linear interdependencies among multiple time series. It extends the autoregression (AR) model by allowing for more than one evolving variable, which is treated as endogenous—that is, each variable in the system is modelled as a linear combination of past values of itself and past values of all other variables in the system. Endogenous variables within a Vector Autoregression (VAR) model are those variables that are explained within the system of equations. Each endogenous variable is modelled as a function of the lagged values of itself and the lagged values of all other endogenous variables in the system. While traditional VAR models treat all variables as endogenous, certain specifications also allow the handling of exogenous variables, which influence the endogenous variables but are not themselves explained within the model.

2.2 Data Exploration and Preprocessing

2.2.1 Ethnicity-based Obesity Dataset for Students (Grades 9 -12)

2.2.1.1 Data & Preprocessing

Our analysis leveraged two main datasets: the "Nutrition, Physical Activity, and Obesity Data" from the CDC, and the "Meat Production Data," encompassing a bi-yearly time interval from 2007 to 2019. These datasets were instrumental in understanding broader dietary trends alongside soda consumption patterns and their effects on obesity rates prevalent for American teens.

Our preprocessing steps were designed to streamline the datasets for analysis, focusing on retaining relevant variables and structuring the data to fit a Panel Vector Autoregression (PanelVAR) model. For the nutrition and obesity dataset, we removed a series of redundant columns that did not contribute to the analysis of soda consumption rates and obesity rates to reduce dimensionality for PanelVAR, such as alternate data values, class identifiers, data source types, and various stratification identifiers. This simplification was crucial for focusing on the most impactful variables, like state, year, and obesity rates.

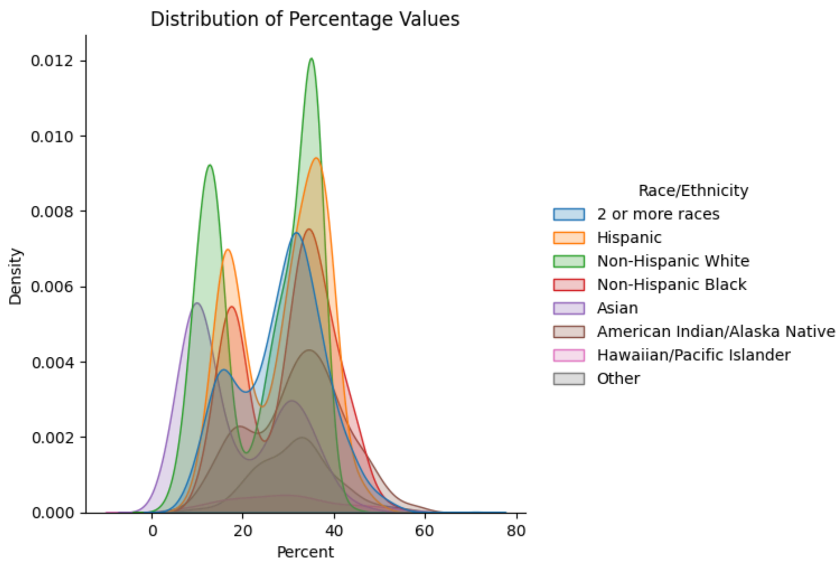


Figure 2.2.1: Probability Density Distribution of a racial group being obese/overweight

Furthermore, the "Nutrition, Physical Activity, and Obesity" dataset was stratified by year, state, and race/ethnicity to create a panel dataset, with each panel being a State-Race pair. The former stratification was used to account for geographical and policy variations and, as displayed by Figure 2.2.1, the latter accounts for cultural factors influencing soda consumption and obesity. This stratification strategy allowed us to create a panel dataset consisting of 897 observations spanning 204 different panels conducive to our PanelVAR analysis and enabling us to account for fixed effects and temporal dynamics.

		Before Seasonal Differencing	After Seasonal Differencing
	ADF Statistic	-1.616888556	-10.2826139
	p-Value	0.474360881	3.74531E-18
Critical Values at Significance Levels	1%	-3.499636534	-3.500378887
	5%	-2.891830773	-2.892151967
	10%	-2.582928338	-2.583099796

Figure 2.2.2: ADF Test Statistic (*1%, 5%, 10%*) and p-Value scores for Meat Production

Given the time series nature of the meat production dataset, we conducted Augmented Dickey-Fuller (ADF) stationarity tests to assess the time series properties of the data. Discovering non-stationarity in the series, we applied seasonal differencing to achieve stationarity, an essential step before incorporating these variables into the PanelVAR model. To further explore this relationship, we plotted a seasonal decomposition curve for the meat production confirming our ADF test results as the data exhibited a clear seasonal pattern.

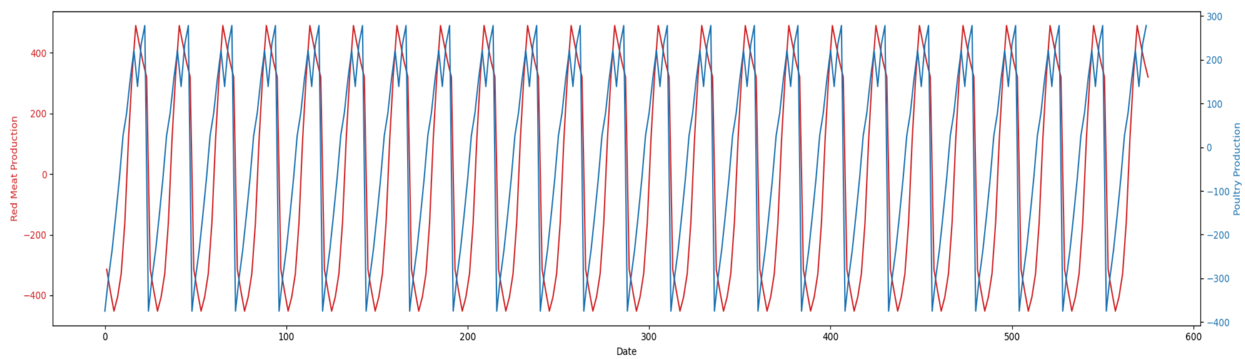
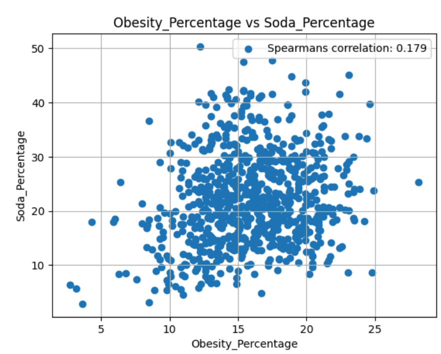


Figure 2.2.3: Seasonality graph of Meat Production paired by *Red Meat (Red) & Poultry (Blue)*

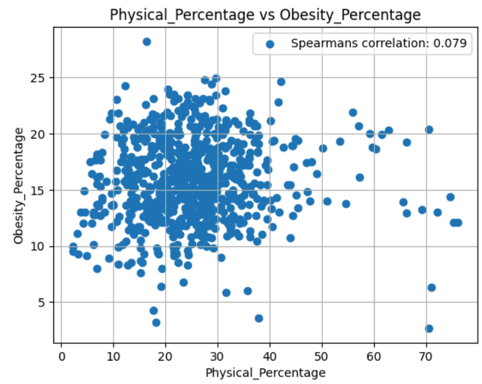
We wanted to study the impact of soda consumption on obesity rates for teenagers, after controlling for potential endogeneity and reverse causality. As true for many time series relationships, the link between soda consumption and obesity is potentially bidirectional, with the soda consumption manifesting itself in increased obesity with a certain lag, and higher obesity rates changing consumer behaviour, potentially impacting soda consumption. Moreover, higher obesity rates in a period may correlate with increased soda and processed food sales in the past few years, which can encourage soft drink companies to increase production in the subsequent years. To model these bidirectional relationships, we decided to proceed with a Panel Vector Autoregression Model, a multivariate time series modelling technique which can treat both our variables of interest as endogenous in the model, while exploiting the cross-sectional aspect of nutritional and physical activity datasets to make more robust causal inference.



To augment our analysis, we also included meat production data as an exogenous factor (exact details explained in the subsequent section), considering its relevance to dietary patterns and potential impact on obesity. This dataset was merged with our main panel dataset, applying 3 lag intervals to the meat production variable to capture its delayed effect on obesity rates. We decided to go with only two lags in the final model, based on BIC results.

Our preliminary findings suggested the existence of a relationship between soda consumption rates and obesity rates, with notable variations across different states and demographic groups. This relationship was strengthened by the existence of a $\rho \approx 0.179$ Spearman's correlation score which made it a worthwhile relationship to explore.

Furthermore, we also considered physical activity to be used as another control since it showed a low correlation with obesity rates and did not appear to be logically linked to soda consumption. This was done in the spirit of parsimony, in order to avoid inferring a large number of parameters from limited data.



In the subsequent sections of our analysis, we will delve deeper into the specific impacts of soda consumption on obesity rates, controlling for various confounding factors.

2.2.1.2 Model Specification and Methodology

$$Z_{it} = \beta Z_{it-1} + \gamma_1 X_{t-1} + \gamma X_{t-2} + f_i + e_{it}$$

Z is the vector {Soda_Percentage, Obesity_Percentage} of endogenous variables, X_{t-1} is the variable meat_prod_diff, which is the first differenced value of meat production at lag 1, X_{t-2} is the first differenced value of meat production at lag 2, and f_i denotes the fixed effects. After conducting the regression and evaluating the Bayesian Information Criterion (BIC) results, we decided to take a lag length of 1, which aligns with the fact that the survey has a biyearly frequency, doing away with the need for longer lags.

Extending the standard Vector Autoregression (VAR) model to a panel data framework offers several benefits, including increasing the robustness of findings by adding a cross-sectional dimension to the survey data, which had a short time frame. Each cross-sectional unit was created by grouping the sample based on state and race. This allowed us to take into account the heterogeneity caused by geographical, policy, and cultural differences. Panel VAR also allows the inclusion of exogenous regressors, which enables us to include nationwide factors like meat production as controls since it is correlated to both our variables of interest. In particular, we included first-differenced lagged values of meat production to control for non-stationarity and endogeneity.

To overcome individual heterogeneity among cross-sectional units, we had to account for fixed effects or the time-variant characteristics across different cross-sectional units. The mean-differencing method, typically employed to remove fixed effects, would have resulted in biased coefficients since the fixed effects are likely to be correlated with the lagged values of the dependent variables. To circumvent this issue, we adopt forward mean-differencing, known as the 'Helmert procedure' or forward orthogonal deviations (Arellano & Bover, 1995). Forward mean differencing can also help tackle any non-stationarity in the data. Moreover, since this approach does not disturb the orthogonality between a variable and its lagged values, we can use System GMM, a Generalized Method of Moments technique that uses the lagged values of the variables as instruments.

After running the VAR Model, we plotted Orthogonalized Impulse Response Functions (OIRF) and Forward Error Variance Decompositions to study feedback loops and analyze the variations in one variable explained by the other in bootstrapped forecasts.

2.2.1.3 Results

	Soda_Percentage	Obesity_Percentage
lag1_Soda_Percentage	0.7191 ***	0.1801 ***
	(0.0478)	(0.0380)
lag1_Obesity_Percentage	0.1411 *	0.7859 ***
	(0.0567)	(0.0464)
Meat_prod_lag1	-0.0082 ***	0.0050 ***
	(0.0019)	(0.0014)
Meat_prod_lag2	-0.0079 ***	0.0022
	(0.0013)	(0.0012)

*** p < 0.001; ** p < 0.01; * p < 0.05

Figure 2.2.4: Estimated Coefficients and Standard Errors in a Panel VAR Model

As can be seen from the results, there is a positive bidirectional relationship between soda consumption and obesity rates among teenagers, with the lagged value of each being a determinant of the other at a statistically significant level.

To confirm the robustness of our results, we conducted the Hansen's Test for Overidentification of Features to check for the robustness of our GMM estimates, for which we got a p-value of 0.193, indicating that the instruments used are not serially correlated with the error term, supporting the validity of our GMM instruments.

2.2.1.4 Orthogonalized Impulse Response Functions

Here, we plot the orthogonalized impulse response functions, which show how an orthogonal one percentage point shock to one variable affects the other variable in the system over time, taking into account the contemporaneous interactions among variables, and keeping other factors constant. Here, an orthogonal shock refers to one that is uncorrelated to shocks in other variables.

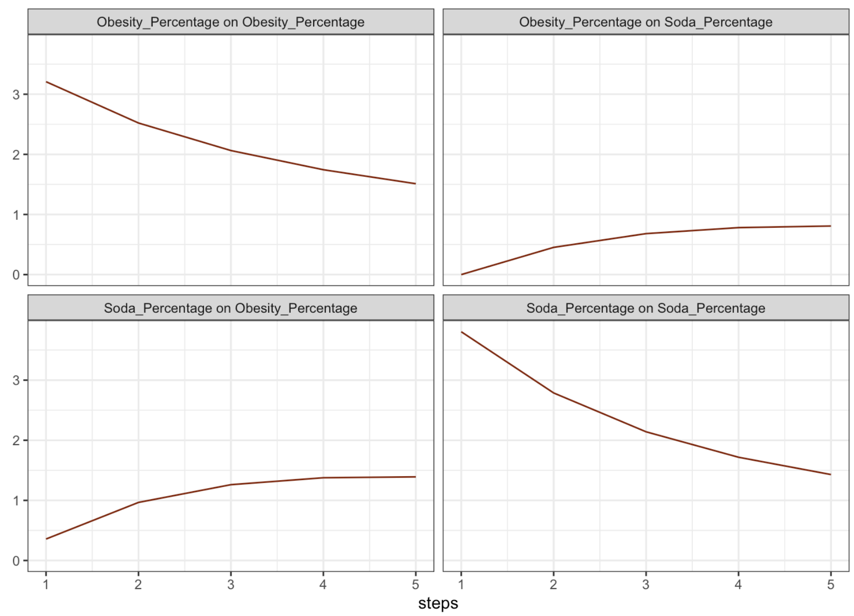


Figure 2.2.5: Orthogonalized Impulse Response Functions for Obesity and Soda Percentages

As can be seen from the graphs, both soda to obesity and obesity to soda form a positive feedback loop, where a shock in one variable leads to a steady increase in the other over time.

2.2.1.5 Forecast Error Variance Decomposition (FEVD)

FEVD quantifies how much of the forecast error variance of each variable is attributable to its own shocks versus shocks to other variables, thus helping identify the primary drivers of the movement within the system.

We can see that the percentage of teenagers who consume soda is able to explain a particularly high degree of variance in obesity percentage. While the percentage of variance in soda consumption explained by the obesity rate is rather low, it is found to be steadily increasing.

```
$Soda_Percentage
     Soda_Percentage Obesity_Percentage
[1,]       1.0000000        0.000000000
[2,]       0.9908652        0.009134815
[3,]       0.9756469        0.024353076
[4,]       0.9587790        0.041221035
[5,]       0.9427228        0.057277186

$Obesity_Percentage
     Soda_Percentage Obesity_Percentage
[1,]      0.01224069          0.9877593
[2,]      0.05988108          0.9401189
[3,]      0.11251689          0.8874831
[4,]      0.15951593          0.8404841
[5,]      0.19808991          0.8019101
```

In conclusion, we are able to identify a positive feedback loop between soda consumption and obesity rates, after accounting for stationarity and controlling for the relevant variables.

2.2.2 Stock and Commodities Dataset for Sugar-Related Brands

2.2.2.1 Aim

Following from our investigation up till now, we now decided to explore the relationship between Soda & Sugary beverage companies with Sugar Prices and Sugar imports. The soda and beverage industry stands as a vital sector within the global economy, with companies such as Coca-Cola (KO) and PepsiCo (PEP) commanding significant market influence. According to The Guardian, Coca-Cola and PepsiCo together account for almost 70% of soft drinks market share in the US, thus we chose to focus our research on these 2 tickers. Understanding the intricate relationships within this industry is crucial for investors, policymakers, and industry stakeholders alike.

This report aims to investigate the dynamic interplay between key variables affecting the performance of soda and beverage companies in the US market. Specifically, we utilize a Vector Autoregression with Exogenous Variables (VARX) approach to analyze the relationships among KO and PEP stock prices, sugar prices, and import levels, while considering exogenous factors such as crude oil price, Consumer Price Index (CPI), retail sales, and a binary variable representing the impact of the COVID-19 pandemic.

2.2.2.2 VARX: Brief Explanation and Justification

The VARX model, or Vector Autoregression with Exogenous Variables, is a statistical tool used to analyze the dynamic relationships between multiple time series variables. In VARX, alongside the lagged values of endogenous variables, exogenous variables are included, providing a more comprehensive understanding of the system under study, and allowing for inclusion of exogenous control variables This approach allows researchers to assess the impact of external factors on the variables of interest, enhancing forecasting accuracy, risk management, and decision-making.

2.2.2.3 Selection of Endogenous and Exogenous Variables

In our research, we have segregated the variables into endogenous and exogenous categories to comprehensively analyze the dynamics of the soda and beverage industry. Endogenous variables, including KO and PEP stock prices and sugar-related metrics such as sugar prices and import levels, are integral components directly influenced by the internal workings of the industry. These variables are essential indicators of the financial performance and operational dynamics of soda and beverage companies.

On the other hand, exogenous variables, such as crude oil prices, CPI, and retail sales, represent external factors that exert influence on the industry but are not determined by it. By including these exogenous variables, we aim to capture the broader economic context and external shocks that impact the soda and beverage sector.

Additionally, the binary COVID variable serves to account for the unprecedented effects of the pandemic on market dynamics, providing further insights into the industry's resilience and adaptation strategies amidst global disruptions. This division enables a holistic understanding of the interplay between internal operations and external influences, facilitating informed decision-making and strategic planning within the soda and beverage industry.

2.2.2.4 Features Sourcing, Cleaning and Engineering

The most basic yet arguably the most important features were the KO and PEP prices, which were obtained using a mixture of the provided datasets and through the YahooFinance Python API. However, just using the prices is not enough due to the variation in the overall price scale. Thus, we engineered features comparable to each other by taking the *log returns* of both KO and PEP, which added the following benefits:

- **Stationarity**: Log returns tend to exhibit stable statistical properties over time, simplifying analysis and forecasting.
- **Normalization**: They provide a standardized measure of price changes, enabling comparisons across assets and periods.
- **Interpretability**: Log returns represent percentage changes, offering intuitive insights into price movements.
- **Symmetry**: They capture both positive and negative returns consistently, facilitating unbiased analysis and modelling.

After calculating the log returns for KO and PEP, we further engineered *Excess Returns*, which were computed by taking the differences in KO/PEP log returns with the log-returns of the SNY. By subtracting the log-returns of the benchmark from those of KO and PEP stocks, we isolate the portion of their returns that cannot be attributed to broader market movements. This enables us to discern the unique performance of KO and PEP relative to the market, shedding light on their idiosyncratic movements.

Next, we prepared the *Sugar Price* to input it as an endogenous variable. Similar to the stock price data, we simply took the *log-returns* of the Sugar Price as well, offering benefits including normalization and stationarity, as well as the ease of direct comparison with the Excess returns.

Additionally, we then prepared data for *US Sugar Import*. The reason for choosing this is that this acts as an effective proxy for the Demand and Supply of Sugar, with the US not being the main producer of Sugar. This data was not provided directly for the Datathon, thus we sourced it from the Federal Reserve Economic Data (FRED) database. Up till now, the time intervals we were dealing with for the Sugar Price and Excess Returns were monthly. However, there was limited monthly data available for US Sugar Import, dating back to only a few years. Thus we choose to use the limited available monthly data to find the seasonality in the dataset so that we can potentially use it to convert yearly data (which was available for a longer time frame) into monthly data.

Using seasonal decomposition, we were able to arrive at the following:

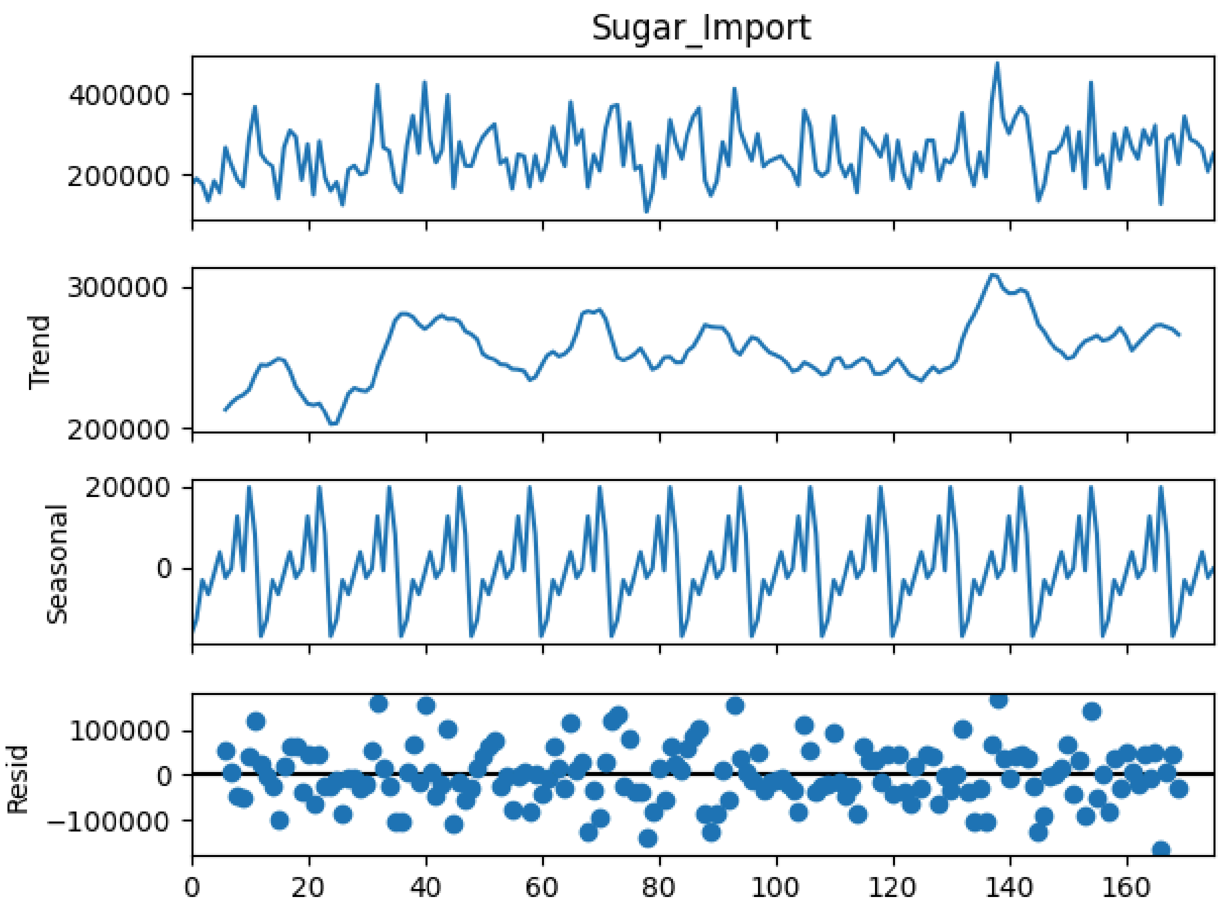


Figure 2.2.6: Looking at the Trend and Seasonal Component of US Monthly Sugar Import Data

The graph above shows that there is very strong seasonality in Sugar Imports, which was used to calculate the monthly ratios to potentially extend monthly US import data by converting yearly data without too much error.

Moreover, as part of our EDA, we ran Granger Causality tests between Sugar Imports data and Sugar Prices data. Granger Causality is a statistical method used to determine if a one-time series variable can help predict another. In our experiment, the tests revealed compelling evidence of a causal relationship between Sugar Imports and Sugar Prices. Specifically, as the number of lags increased, the statistical significance of this relationship became increasingly pronounced. At four lags, all tests indicated highly significant results at the 1% level, suggesting a robust causal relationship between the variables. This finding underscores the importance of considering both Sugar Imports and Sugar Prices as endogenous variables in our overall investigation. By incorporating both variables, we can capture the dynamic interplay between them and gain a deeper understanding of the factors influencing sugar prices, enabling more informed analysis, forecasting, and decision-making in relevant industries.

```
Granger Causality
number of lags (no zero) 2
ssr based F test:         F=3.4962  , p=0.0325  , df_denom=168, df_num=2
ssr based chi2 test:   chi2=7.2005  , p=0.0273  , df=2
likelihood ratio test: chi2=7.0547  , p=0.0294  , df=2
parameter F test:         F=3.4962  , p=0.0325  , df_denom=168, df_num=2

Granger Causality
number of lags (no zero) 3
ssr based F test:         F=4.0638  , p=0.0081  , df_denom=165, df_num=3
ssr based chi2 test:   chi2=12.7086 , p=0.0053  , df=3
likelihood ratio test: chi2=12.2610 , p=0.0065  , df=3
parameter F test:         F=4.0638  , p=0.0081  , df_denom=165, df_num=3

Granger Causality
number of lags (no zero) 4
ssr based F test:         F=4.4157  , p=0.0021  , df_denom=162, df_num=4
ssr based chi2 test:   chi2=18.6440 , p=0.0009  , df=4
likelihood ratio test: chi2=17.6960 , p=0.0014  , df=4
parameter F test:         F=4.4157  , p=0.0021  , df_denom=162, df_num=4

Granger Causality
number of lags (no zero) 5
ssr based F test:         F=3.4628  , p=0.0054  , df_denom=159, df_num=5
ssr based chi2 test:   chi2=18.5119 , p=0.0024  , df=5
likelihood ratio test: chi2=17.5716 , p=0.0035  , df=5
parameter F test:         F=3.4628  , p=0.0054  , df_denom=159, df_num=5
```

Figure 2.2.7: Granger Causality Results for Sugar Prices and US Sugar Imports

Furthermore, the dataset titled `exog_final` encapsulates a multifaceted exploration of economic indicators and their interrelations, particularly emphasizing the period marked by the COVID-19 pandemic. Derived from the robust repositories of the Federal Reserve Economic Data (FRED), this dataset amalgamates critical exogenous features instrumental for analyzing economic trends and shocks.

Central to our dataset is the inclusion of Crude Oil Commodity Prices. The rationale behind this selection is twofold: firstly, crude oil serves as a fundamental input across diverse sectors, thus acting as a bellwether for manufacturing costs. Secondly, empirical evidence underscores a pronounced correlation between crude oil prices and the pricing dynamics of other commodities, such as corn, which is pivotal to the confectionery and fast food industry. This correlation provides a lens through which the ripple effects of oil price fluctuations on broader economic sectors can be discerned.

Further enriching this dataset are variables such as the Consumer Price Index (CPI) Growth Rate, Retail Sales, and a Binary Variable for COVID-19 Disruptions. The CPI Growth Rate is scrutinized to unearth any deviations in the established correlation patterns during the tumultuous COVID phase, thereby offering insights into inflationary trends. Retail Sales data is incorporated as a proxy for economic growth, reflecting consumer spending behaviours in varying economic climates.

Lastly, a binary variable was added. This variable delineates the COVID-19 disruption epoch (March 2020 to January 2022), and serves as a pivotal tool for demarcating the pandemic's temporal impact. This inclusion allows for a nuanced analysis of the pandemic's immediate effects on economic activities and the efficacy of subsequent recovery measures

Feature Name	Nature (Endogenous/Exogenous)	Status (Cleaned/Engineered/Sourced)	Source
KO & PEP Close	Endogenous	Sourced	Yahoo Finance
KO & PEP Log Returns	Endogenous	Engineered	-
Excess Returns	Endogenous	Engineered	-
Sugar Price	Endogenous	Sourced	Provided
Sugar Price Log Returns	Endogenous	Engineered	-
Monthly US Sugar Production Data	Endogenous	Sourced	FRED
Seasonal/ Monthly Ratios for US Sugar Production Data	-	Engineered	-
Crude Oil Prices (Crude_Oil_Price)	Exogenous	Sourced	FRED
Consumer Price Index Growth Rate (CPI_Growth_Rate)	Exogenous	Sourced	FRED
Retail Sales (RSXFS)	Exogenous	Sourced	FRED
Binary COVID Variable (COVID_Var)	Exogenous	Engineered	-

Figure 2.2.8: Table of Endogenous and Exogenous Variables utilized in VARMAX

2.2.2.5 Results & Limitations

Through rigorous modelling of endogeneity and inclusion of relevant control variables, we found that there is not statistically significant link between log returns of the stocks of Carbonated Companies, sugar prices and sugar imports.

A few possible explanations for this includes:

- Sugar is only a small component of their overall costs structure, thus leaving unaffected by all except the most major market fluctuations
- Investors and shareholders may not penalize these mega-coorperations based on short-term prices and profit fluctuations, given their long term stability.
- Coca-cola and Pepsico may employ quantitative techniques, including price hedging measures, which may protect them from short term fluctuations.

Results for equation Excess_Return

	coef	std err	z	P>|z|	[0.025	0.975]
intercept	-0.0397	0.464	-0.085	0.932	-0.949	0.870
L1.Excess_Return	-0.1744	0.409	-0.426	0.670	-0.976	0.628
L1.Sugar_Log_Return	0.0077	0.250	0.031	0.975	-0.482	0.498
L1.Sugar_Import	5.344e-10	8.47e-08	0.006	0.995	-1.65e-07	1.67e-07
L1.e(Excess_Return)	-0.0242	0.417	-0.058	0.954	-0.841	0.793
L1.e(Sugar_Log_Return)	-0.0054	0.266	-0.020	0.984	-0.526	0.515
L1.e(Sugar_Import)	-5.325e-08	8.47e-08	-0.629	0.530	-2.19e-07	1.13e-07
beta.CPI_Growth_Rate	0.0100	0.012	0.862	0.389	-0.013	0.033
beta.COVID_Var	0.0013	0.012	0.110	0.912	-0.022	0.025
beta.RSXFS_log	0.0013	0.017	0.077	0.939	-0.033	0.035
beta.Crude_Oil_Price_log_return	-0.1889	0.037	-5.077	0.000	-0.262	-0.116

Figure 2.2.8: Results for equation of Excess Return

3 Conclusion and Limitations

3.1 Conclusions and Future Directions

In this comprehensive analysis, our findings highlight a statistically significant bidirectional relationship between soda consumption and obesity rates across various states and racial demographics in the United States, as well as the nuanced interplay between sugar prices, sugar imports, and the carbonated beverages industry. Employing a Panel Vector Autoregression (PanelVAR) model, we've captured the dynamic interaction between soda consumption and obesity rates, enriched by the inclusion of meat production data as an exogenous variable, to reflect a broader dietary perspective.

Our analysis reveals that soda consumption not only forecasts but also results from rising obesity rates among teenagers, creating a self-reinforcing cycle that exacerbates obesity prevalence. This relationship, underscored by the Orthogonalized Impulse Response Functions and the Forecast Error Variance Decomposition (FEVD), identifies soda consumption as a significant contributor to the variance in obesity rates, presenting a critical opportunity for public health interventions.

Moreover, our exploration into the economic aspects of the carbonated beverages industry unveils how sugar prices and imports influence production and pricing strategies within the industry, subsequently affecting soda consumption patterns. This economic analysis provides a broader context, showing how global trade policies and economic trends can indirectly influence dietary choices and, by extension, obesity rates. The interconnectedness of soda consumption's impact on obesity and the economic forces shaping the carbonated beverage market underscores the complexity of tackling the obesity epidemic.

These insights offer valuable guidance for policymakers and health advocates, suggesting that reducing soda consumption among adolescents could significantly impact the fight against obesity. Additionally, the implications of our findings for investors and stakeholders in the food and beverage industry highlight the importance of considering economic trends and public health initiatives in strategic decision-making.

Future research should build on this foundation by using longitudinal data to observe changes over time and by including more variables that might affect or be affected by soda consumption and obesity rates. Applying advanced machine learning techniques could also enhance our ability to predict trends and dissect the intricate relationships uncovered in this study, providing a more thorough understanding of the factors driving the obesity epidemic.

3.2 Limitations

However, the robustness of our findings is bound by certain limitations. Firstly, the sample size and granularity of data, especially pertaining to soda consumption and obesity rates, limit the generalizability of our results. Larger datasets could offer a more nuanced understanding and validate the robustness of our findings across broader populations.

Secondly, the endogenous nature of the PanelVAR model requires careful consideration of potential reverse causality between the variables, which, despite our methodological precautions, could still present confounding effects. The interdependence of soda consumption, obesity rates, and meat production is complex and may be influenced by factors not captured in our analysis.

Moreover, our feature engineering, while sufficient for the scope of this analysis, could be improved by incorporating higher-dimensional embeddings. Such embeddings might capture additional attributes of dietary behaviours and their impact on obesity, like named-entity recognition, that were not considered in this analysis.

Lastly, temporal dynamics represent another limitation. Our study focuses on a snapshot in time, and the relationships observed may shift with changing cultural norms, regulatory policies, and industry practices. The pace at which these variables evolve necessitates continuous monitoring and updated analysis to ensure the relevance of the findings.